Axles and alignment

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements and observations to improve straight rolling.

SUCCESS CRITERIA - I can...

1. Check axle parallelism and wheel clearance.
2. Record a repeatable alignment test.
3. Use measurements or observations to choose one adjustment.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “axle” and “alignment” to explain your answer. Sentence starters are optional.

axle: a shaft associated with a rotating wheel or gear

alignment: how the directions or positions of parts line up

clearance: space that lets neighboring parts avoid rubbing or colliding

01 Warm-up: bring back an earlier idea

A **50**-turn coil has flux per turn change by **0.007 Wb** in **0.5 s**. **Find average induced-emf magnitude**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: clearance = available gap - part width.
Use positive clearance and **check free rotation**. Alignment and wheel rubbing are separate checks; change one feature at a time.

Worked reasoning: Parallel axles can still cause drag if wheels rub the chassis. Alignment and clearance are separate checks; a straight top view does not verify free rotation.

Your turn: Draw a top view and a side detail marking an axle direction and a wheel gap.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

Axles and alignment

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

On your top-view sketch, mark both axle lines. **Add a check** for parallelism and **show clearance** between each wheel and the chassis.

Sentence starter: In my work, I used … to show … . I labeled it … .



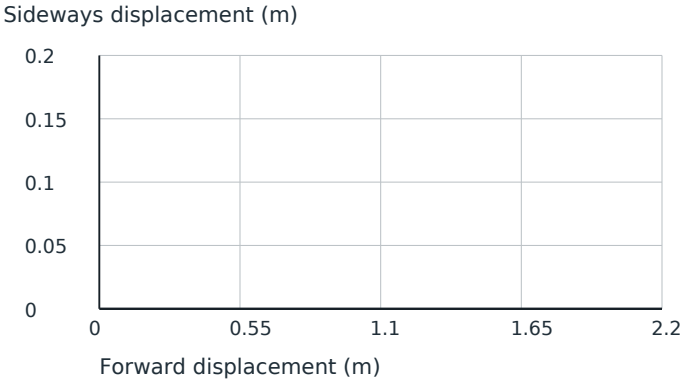
Marzano check - circle: 1 2 3

04 Apply the physics

A sample straight-tracking test ends **2.0 m** forward and **0.12 m** right of the starting point. Plot the endpoint.
1. **Calculate the displacement magnitude** and direction using $\tan^{-1}(\mathbf{0.12/2.0})$.
2. **Explain what these measurements say** about straight tracking.

Word help: magnitude = size of a quantity without its direction

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: 1 2 3

Axles and alignment

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A car veers right once. Does that prove the rear axle is crooked? **Name another possible cause** and a discriminating check.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

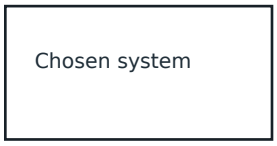
06 Review: connect an earlier idea

A **4 Ω** resistor carries **6 A**. **Find power** using both IV and I^2R, showing the implied voltage.
Label input, useful output and other energy transfers.

Word help: voltage = electric potential difference; energy transferred per charge

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Two designs each complete three identical free-roll tests.
- Before adjustment: **0.18**, **0.22**, and **0.20 m** right.
- After adjustment: **0.04**, **0.06**, and **0.05 m** right.
1. **Compare the mean offset** and **range**.
2. **Explain why these results tell** you more than one excellent run.

Sentence starter: Both … . They differ in … because … .

Trial	Before offset (m)	After offset (m)
1	0.18	0.04
2	0.22	0.06
3	0.2	0.05

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Repeated **3 m** runs finish **0.20**, **0.24**, **0.22 m** left of the line. **Find mean deviation** and range; design a controlled adjustment and a fair comparison after it.

Sentence starter: My design includes … . It does what is needed by … .



Marzano check - circle: 1 2 3